

07 · Did the regional campaign work? — synthetic control (Anchor B)

July 13, 2026

The business decision. We ran a TV campaign in one metro area — a **DMA** (*Designated Market Area*, the standard US media-buying region) — and left the rest of the country dark. Sales in that metro are up since launch. Before we spend to roll the campaign out nationally: **was the lift caused by the campaign, or would sales have risen there anyway?**

Why this is hard: there was no control group

Unlike the uplift notebook (01), here there is **no randomized hold-out** — we didn’t flip a coin to decide which metro got the campaign, we just picked one. So we can’t simply compare it to “the others.” And we can’t just compare the treated metro’s sales *after* vs *before*, because sales everywhere are pushed around by a shared **trend** (secular growth), **seasonality**, and a common **macro wave** (the economy, category demand). A naive before/after would credit the campaign with an uplift the market would have delivered on its own. We need the one thing we can never observe: **what sales in that metro would have been if we had not run the campaign** — its *counterfactual*.

The idea: build the counterfactual from the other markets

Synthetic control (Abadie & Gardeazabal, 2003) constructs that missing counterfactual as a **weighted blend of the untreated markets** — the “**donor pool**.” It searches for a set of weights so that the blend closely tracks the treated market’s sales *in the pre-launch period*. If a weighted combination of, say, 60% Denver + 25% Phoenix + 15% others matched our metro week-for-week *before* the campaign, then — assuming those markets keep moving together — that same blend is our best guess of what our metro *would* have done *after*, absent the campaign. The **gap** between the metro’s actual sales and this “synthetic” version, after launch, is the estimated lift.

The reason it can work at all: all markets share the same underlying **factors** (trend, season, macro). A blend of donors that matches the treated unit’s pre-period is, in effect, matching it on those shared factors — so the blend and the treated unit stay locked together *unless something (the campaign) breaks them apart*.

We do this the Bayesian way, so instead of one number we get a **posterior** — a full distribution of plausible lifts — which gives us honest uncertainty, a real significance test (a **placebo permutation**, defined in Depth B), and robustness checks a VP of Growth can interrogate before committing national budget.

How this notebook is organised

The **7-step contract** (question · simulate a known truth · identify · estimate · validate · decide in € · caveats) plus the full **three depths**:

- **Depth A** — why the tempting naive estimators fail (a bake-off)
- **Depth B** — placebo inference in *space* and *time*, with the scale-free Abadie RMSE-ratio test
- **Depth C** — the euro rollout decision and its sensitivity to assumptions

```
FAST=False SC sampling={'draws': 1500, 'tune': 1500, 'chains': 4}
```

2 · Simulate a ground truth (why fake data first)

As in every notebook here, we **cannot grade a causal method on real data** — the counterfactual (“what our metro would have done without the campaign”) is never observed. So we build a simulator where we *plant* the true lift, confirm the method recovers it, and only then trust it.

The simulator produces a weekly **sales panel**: a table with one row per market per week, for 30 DMAs over 60 weeks. Each market’s sales are built from **shared latent factors** — a rising trend, a seasonal cycle, and a random macro wave common to all markets — combined with that market’s own baseline level, its own sensitivity to each factor (“factor loadings”), and idiosyncratic noise. The campaign launches in **week 40** in market 0 only, adding a **true +12% lift** from then on.

The shared macro wave is the villain: it moves *every* market together, so it’s exactly what a naive before/after would mistake for campaign effect. But it’s also the hero of the method — because the factors are *shared*, a weighted blend of the *other* markets can reconstruct market 0’s untreated path. That is precisely what makes the effect *identifiable* here (recoverable in principle), and the plot below shows the setup: one dark line (treated) inside a bundle of grey donors, all riding the same wave.

The data-generating model — exactly what `dgp.geo_panel` implements (defaults & seed in `src/cmp/dgp.py`). Markets $i = 0, \dots, 29$, weeks $t = 0, \dots, 59$. Three **shared latent factors**:

$$\begin{aligned} g_t &= 0.4 t && \text{(trend),} \\ s_t &= 8 \sin\left(\frac{2\pi t}{26}\right) + 4 \sin\left(\frac{2\pi t}{13}\right) && \text{(season),} \\ m_t &= \sum_{u \leq t} \eta_u, \quad \eta_u \sim \mathcal{N}(0, 1.2^2) && \text{(macro random walk).} \end{aligned}$$

Each market has its own baseline $c_i \sim U(80, 140)$ and factor loadings $\lambda_i = (\lambda_{i1}, \lambda_{i2}, \lambda_{i3})$, each $\sim U(0.6, 1.4)$:

$$Y_{it} = c_i + \lambda_{i1} g_t + \lambda_{i2} s_t + \lambda_{i3} m_t + \varepsilon_{it}, \quad \varepsilon_{it} \sim \mathcal{N}(0, 3^2),$$

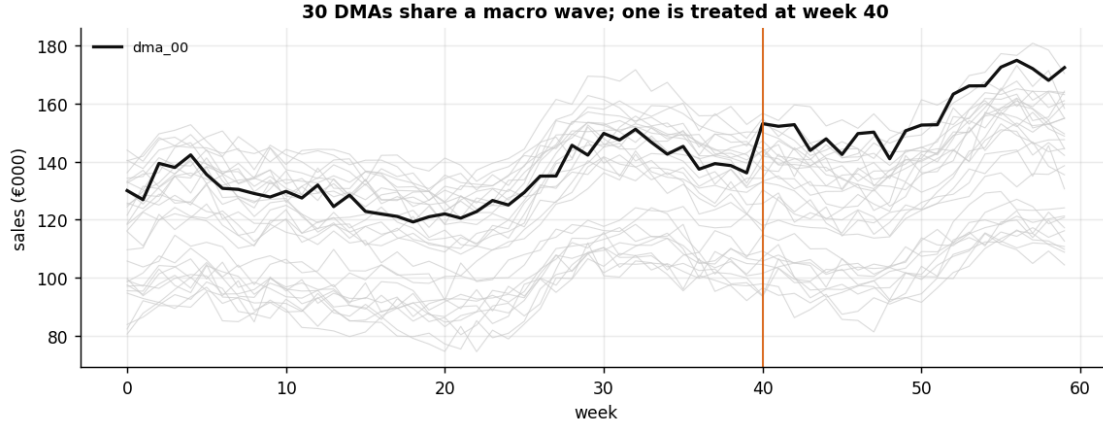
and the campaign adds a **proportional lift to market 0 only**, from week 40:

$$Y_{0t} = c_0 + \lambda_{01} g_t + \lambda_{02} s_t + \lambda_{03} m_t + \varepsilon_{0t} + \Delta_t, \quad \Delta_t = 0.12 (c_0 + \lambda_{01} g_t + \lambda_{02} s_t + \lambda_{03} m_t) \mathbf{1}[t \geq 40].$$

This factor structure is precisely why synthetic control works here: every market rides the *same* (g_t, s_t, m_t) , only with different loadings, so a convex combination of donors whose blended loadings

match $(\lambda_{01}, \lambda_{02}, \lambda_{03})$ reproduces market 0's untreated path. It's also why the naive estimators fail — the random-walk m_t moves all markets together, and a before/after comparison books that shared drift as “lift.”

Treated: dma_00, launch week 40. TRUE lift €17.0k/week, total €339k.



3 · Identify — the estimand and the assumptions it rests on

The estimand (the precise thing we *want*). For each post-launch week t , the causal effect on the treated market is the potential-outcome contrast

$$\tau_t = Y_{0,t}(1) - Y_{0,t}(0), \quad t > T_0,$$

where $Y_{0,t}(1)$ is **observed** (the campaign ran) but $Y_{0,t}(0)$ — what week- t sales *would have been* with no campaign — is the **missing counterfactual**. (Throughout we label the treated unit **market 0** — matching §2's DGP and §4's fit — so the subscript is a *unit* index, not a treatment flag.)

The estimator. Synthetic control fills in that counterfactual with a weighted average of donor markets, so the effect we actually compute is

$$\hat{\tau}_t = Y_{0,t} - \underbrace{\sum_j w_j Y_{j,t}}_{\widehat{Y_{0,t}(0)} = \text{synthetic control}}, \quad t > T_0,$$

where $Y_{0,t}$ is the **treated** market's sales in week t , $Y_{j,t}$ is **donor** market j 's sales, T_0 is the launch week, and w_j are the **donor weights**. The weights are chosen to match the treated market's *pre-launch* path and are constrained to the **simplex**: each $w_j \geq 0$ and they sum to 1 ($\sum_j w_j = 1$). Think of the simplex constraint as “the synthetic market must be a genuine *weighted average* of real markets” — no negative weights, no scaling beyond the observed range.

Why the simplex matters (the core of Abadie's method). If instead we let weights be any numbers (ordinary regression), the “synthetic” market could sit *outside* the range of any real market — **extrapolation** dressed up as a match, which fits the pre-period beautifully and then

diverges wildly after. Forcing weights onto the simplex keeps the synthetic inside the **convex hull** of the donors (loosely, “inside the cloud of real markets”), which is what makes the post-launch projection credible.

The assumptions — each gets an explicit check later:

assumption	plain meaning	checked in
convex hull / no extrapolation	the treated market lies inside the donor cloud; simplex weights enforce it	weights (Step 4)
good pre-fit	the synthetic actually tracks the treated market <i>before</i> launch	pre-RMSE gate (Step 5)
no anticipation	nothing shifts sales <i>before</i> launch (no leaked campaign, no pre-buying)	placebo-in-time (Depth B)
stable factor structure	donors and treated keep co-moving <i>after</i> launch (the loadings matched pre-launch still hold)	long pre-period + placebo-in-time
no spillover / interference	the campaign in market 0 doesn’t change donor markets (SUTVA — the Stable Unit Treatment Value Assumption — for geos)	stress-tested by simulation (Depth B): planted spillover <i>attenuates</i> the estimate, and excluding the contaminated donors restores it — placebo-in-space <i>cannot</i> detect spillover (it contaminates every placebo too), so on real data exclude adjacent / media-overlapping donors up front
no concurrent treated shock	nothing else hit market 0 exactly at launch	untestable — institutional knowledge only

Step 0 · The classical read (no likelihood, no priors, no sampler)

Before any Bayesian machinery, ask what a competent analyst does here **without** it. The answer is not a different analysis: it is the **same estimand** (§3’s per-week gap τ_t , and its 20-week total), identified by the **same assumptions** (§3’s table — convex hull, good pre-fit, no anticipation, no spillover). The causal work lives in the identification, not in the machinery. In this notebook that is unusually easy to see, because the classical estimator *is* the method: Abadie & Gardeazabal’s synthetic control was born frequentist, and §4’s Dirichlet model is its Bayesian re-housing, not a rival.

The estimator, naked. Pick the donor weights that best reproduce the treated market’s *pre-launch* path, subject to §3’s simplex constraint:

$$\hat{w} = \arg \min_{w \in \Delta^{J-1}} \sum_{t < T_0} \left(Y_{0t} - \sum_{j=1}^J w_j Y_{jt} \right)^2, \quad \Delta^{J-1} = \left\{ w : w_j \geq 0, \sum_j w_j = 1 \right\}.$$

That is constrained least squares — a quadratic objective on a convex set — solved numerically by **SLSQP** (`scipy`’s *Sequential Least-Squares Quadratic Programming* optimizer; wrapped as `est.sc_weights_slsqp`). It is **deterministic**, runs in milliseconds, and returns one number per donor. The counterfactual and the effect then follow by plug-in — no further estimation:

$$\widehat{Y_{0t}(0)} = \sum_j \hat{w}_j Y_{jt}, \quad \hat{\tau}_t = Y_{0t} - \widehat{Y_{0t}(0)}, \quad \widehat{\text{total lift}} = \sum_{t \geq T_0} \hat{\tau}_t.$$

Nothing else is fitted: no variance parameter, no prior, no chain. (This same fitter carries *every* robustness check in the notebook — leave-one-out in §5, the placebos below, the launch-date sweep and the spillover stress test in Depth B/C — precisely because it is cheap and deterministic.)

Now the hard part: where does an interval come from? Nowhere, by the usual route. There is no likelihood, so there is no standard error to read off a fit; and even if we bolted a regression standard error onto the gap, it would be answering the wrong question — standard errors describe variability across many independent treated units, and we have **exactly one**. Abadie’s own answer, and the inferential heart of the method, is **placebo (permutation) inference**:

Under the null “the campaign did nothing to market 0”, market 0 is *just another market*. So refit the identical estimator on each donor **as if it had been treated**, collect the post-launch gaps those placebo fits produce — that is the distribution of gaps the world hands you from noise alone — and ask where the real one **ranks**.

Two refinements, both Abadie’s, both used below:

- **Discard placebos with a poor pre-fit** (here: pre-RMSE more than $5 \times$ the treated unit’s). A donor the synthetic cannot match *before* launch produces a meaningless gap that would only pollute the reference distribution.
- **Rank on the RMSE *ratio*, not the raw gap**: post-period RMSE $\div$ pre-period RMSE. It is **scale-free** (fair across big and small markets) and refuses to reward a placebo that simply fits badly *everywhere* (high post **and** high pre $\rightarrow$ ratio ≈ 1).

The **p-value** is the treated market’s rank in that ratio distribution. And an *interval* can be recovered from the very same null, by **inversion**: the placebo totals map out how far the estimator’s 20-week total wanders when the true effect is zero, so subtracting that distribution’s 5th and 95th percentiles from our estimate gives a 90% **randomization interval** for the lift (valid under a constant-effect null — we borrow the placebos’ *spread*, not their location). It is a genuinely design-based interval: it never assumes a shape for the errors, it *measures* it.

Depth B opens the placebo box in full — the whole spaghetti of placebo gaps, how the verdict moves with that $5 \times$ filter, and the mirror test (placebo-in-time). Step 0 needs only its headline number.

CLASSICAL SYNTHETIC CONTROL - SLSQP simplex weights, fit on weeks 0-39

29 donors: 10 carry >1% of the weight; 5.8 effective donors ($1/\text{Sigmax}^2$)

pre-fit RMSE €2.44k - the gate: the synthetic must track the

treated market BEFORE launch, or the post gap means nothing

lift €15.9k/week * total €318k over 20 post-launch weeks
GRADE vs the planted €339k total: off by €-22k (-6%)

INFERENCE - no likelihood, so no standard error. Rank the real gap
against the placebo null instead:

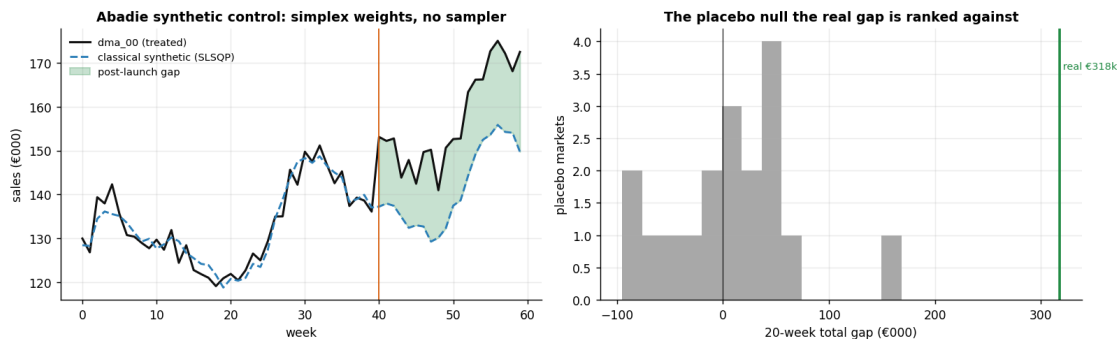
Abadie RMSE-ratio $p = 0.033$ - treated ranks 1 of 30 markets (ratio 6.7
vs a placebo median of 1.3)

raw-gap permutation $p = 0.053$ - 18 of 29 donors survived the
5x pre-fit filter (Depth B reconciles the two p's)

90% randomization interval on the total (placebo null, inverted):

[€245k, €401k] - the planted €339k falls INSIDE it

What this 90% confidence interval does NOT say: that there is a 90% probability the true effect lies inside it. It is a property of the *procedure* - 90% of intervals built this way would cover the truth across repeated samples. This interval either contains the truth or it does not. To get a probability *about* the effect itself - the thing a go/no-go rule like $P(\text{lift} > \text{cost}) \geq 0.9$ actually needs - you need a posterior. That is what the Bayesian section adds.



Read-out — the classical answer, in business terms. The synthetic market tracks the treated one before launch to within the pre-fit RMSE printed above (a couple of € thousand on weekly sales of €80–170k), which is the *precondition* for reading anything into the post-launch gap — and that gap is the campaign’s estimated lift: the per-week and total euro figures above, graded against the planted truth in the same line. The estimator lands close, and slightly **low** — hold that undershoot, \$5x and \$7 return to it.

Look at the weight count while you are here: classical SC puts real weight on a **handful** of donors — the sparse blend Abadie’s method is famous for, and a genuinely useful business object (“our metro behaves like this specific mix of a few named other metros”). §4’s Dirichlet prior will instead spread weight thinly over *many* donors: more stable out-of-sample, less interpretable, and the reason §5’s leave-one-out chart carries two nearby-but-distinct baselines (“all donors (SLSQP)” and “all donors (Bayesian)”).

What the placebo p-value is. It is a **randomization-inference** p-value: *if the campaign had done nothing, a post-launch jump this extreme — relative to its own pre-fit — would have turned up in about this fraction of markets.* It is a statement about a **null** and about a **procedure**: we ran

the same estimator on markets we know were untreated, and the treated market’s statistic came out at the printed rank.

What it is not — and this is the same guardrail this cookbook states in every notebook, sharpened by the fact that the classical arm here has *no interval at all* by default, only a rank. It is **not** the probability that the campaign had no effect. It is **not** one minus the probability the campaign worked. And it is emphatically **not** the probability that the campaign **paid** — the question §6 has to sign off on, which involves the €300k cost, a number the p-value has never met. The randomization interval we inverted out of the same placebo null is a *confidence* interval and inherits exactly the limits printed above: 90% of intervals built that way cover the truth **across repeated samples** (§5x actually goes and measures that rate, on fresh panels — it is one of the few places in this cookbook where the “repeated samples” story is literally executed), and this particular one either covers the truth or it does not.

So: no likelihood, no prior, no sampler — and already a lift estimate, a falsification test, and an honest interval. **This is the baseline the Bayesian section must justify itself against.** What it cannot produce, at any level of effort, is $P(\text{lift} > \text{cost})$. That quantity does not exist in its vocabulary; it exists in a posterior. §4 builds one — and §5x asks, with numbers, whether it is a *trustworthy* one.

4 · Estimate — a Bayesian synthetic control

We put a **Dirichlet prior** on the donor weights and fit them to the pre-period. (The *Dirichlet* is the natural probability distribution over the simplex — draws from it are automatically non-negative and sum to 1 — so it’s the Bayesian way to say “the weights live on the simplex.”) Fitting returns a **posterior** over the weights, and therefore a posterior over the whole counterfactual path and the lift — Abadie’s constrained least squares, but with uncertainty for free.

The fitted model, in symbols. With Y_{0t} the treated market and Y_{jt} the $J = 29$ donors, fit **on the pre-period only** ($t < 40$):

$$Y_{0t} \sim \mathcal{N}\left(\sum_{j=1}^J w_j Y_{jt}, \sigma^2\right), \quad w \sim \text{Dirichlet}(\mathbf{1}_J), \quad \sigma \sim \text{HalfNormal}(5).$$

$\text{Dirichlet}(\mathbf{1}_J)$ is flat over the simplex — no donor favoured a priori, weights non-negative and summing to 1 by construction. The $\sigma \sim \text{HalfNormal}(5)$ scale is likewise tied to the units of the data: sales are in €000s, weekly levels sit around €80–170k, and the DGP’s idiosyncratic noise has sd €3k — so a prior sd scale of €5k is weakly informative on the residual: wide enough to cover any plausible blend error, tight enough to rule out absurd ones (a €50k weekly residual would mean the “synthetic” tracks nothing). The counterfactual for *every* week is then the posterior-predictive blend $\hat{Y}_{0t}^{(0)} = \sum_j w_j Y_{jt} + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ — the ε_t matters: dropping it (using the mean blend alone) understates the uncertainty band, which is exactly the coverage lesson step 5 quantifies — and the weekly lift is $Y_{0t} - \hat{Y}_{0t}^{(0)}$.

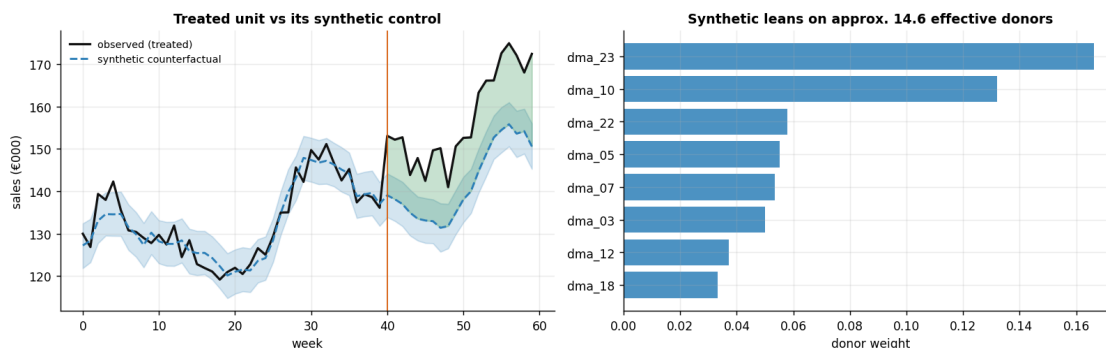
Two things to read off the fit: - **Which donors** the synthetic leans on (the weight bars). One caveat to state plainly: the Dirichlet prior produces a *regularized blend* spread over many donors, rather than classical synthetic control’s sparse handful. That’s **more stable** out-of-sample but **less individually interpretable**. We summarise the spread as an “**effective number of donors**” $= 1/\sum_j w_j^2$ (the inverse Herfindahl index — 1 if all weight is on one donor, J if spread evenly over J donors). - **Pre-fit RMSE** — the root-mean-square gap between the treated market and

its synthetic *before* launch, in €000. Small means the synthetic is a good stand-in; this is the precondition for believing the post-launch gap — Step 5 turns this into a concrete PASS/FAIL gate (pre-RMSE below ~a third of the weekly lift it must detect).

Reading the sampler’s health check. The convergence line printed below reports the three standard MCMC health checks — the numbers that say whether the sampler (the algorithm that draws from the posterior) actually converged. **R-hat** compares the variance within each chain to the variance across chains; 1.00 is perfect and ≤ 1.01 is the usual pass bar. **ESS** (effective sample size) is how many *independent* draws our autocorrelated chains are worth — a few hundred is ample for a posterior mean or interval. **Divergences** are steps where the sampler broke down and silently distrusts that region; **you want 0**. Under this notebook’s FAST teaching profile the chains are deliberately short (2 chains), so R-hat can sit a hair over the 1.01 line and PyMC may print a “problems during sampling” notice — a benign artifact of the small draw count, not a red flag: the ESS below runs well into the thousands and divergences are 0, so the flagship fit is sound (under the FAST profile the ESS is naturally lower but still ample).

On real data. You don’t need our simulator — this method runs on *your own* weekly (or daily) sales-by-market panel, which every company already has: one treated market, the rest as donors. The canonical *public* example is Abadie’s **California Proposition 99** tobacco study (California vs 38 other US states); it has the identical shape (one treated unit, many donors, a known intervention date) and is the standard dataset to reproduce this on.

SC sampling convergence: max r-hat 1.000 - min ESS 3994 - divergences 0
pre-fit RMSE €2.86k * estimated total lift €305k (true €339k) * 90% CI
[€270k, €339k] - at full precision the truth sits just OUTSIDE this band (\$5's
multi-seed check explains why the total-lift band runs optimistic)



5 · Validate — recovery, calibration, the pre-fit gate, and robustness

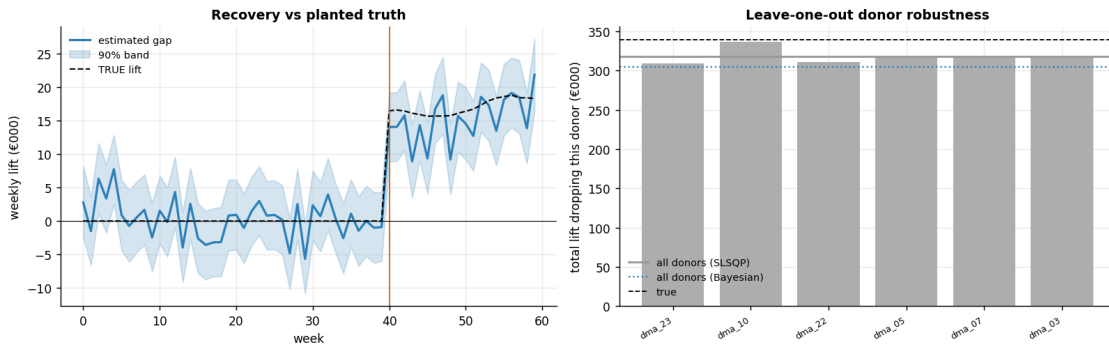
Four checks — and one of them delivers a genuinely humbling result we keep rather than hide:

1. **Recovery & calibration.** The true per-week path should sit inside the posterior band. That band is a **posterior predictive**: it carries not only *which-donors* (weight) uncertainty but the treated market’s own idiosyncratic noise. That noise term is what makes the per-week band near-nominal; a weights-only band is too narrow and can leave the truth *outside* a “90%” interval.

2. **Multi-seed calibration.** One panel is a single draw of the world, so we refit on fresh panels and report the recovery bias and coverage. The honest result: the **per-week** band is near-nominal, but the **total-lift** interval covers well *below* nominal — synthetic control’s genuine sample-to-sample variance is larger than any single panel’s posterior width. So the posterior interval is an *optimistic* guide to the total, and the trustworthy inference is the **placebo permutation test** (design-based, calibrated by construction), not the band. This is *why* the euro call below is “confirm with a second test,” not “the interval says go.” (*The per-seed refits in this check are deliberately coarse 300-draw probes, read only in aggregate; their sampler chatter is silenced.*)
3. **Pre-fit gate** — the pre-period gap must hover around **zero**: concretely, pre-RMSE below about a third of the weekly lift the design must detect (printed as an explicit PASS/FAIL below). A bad synthetic match makes the post gap meaningless — this is the precondition, not a formality.
4. **Leave-one-out donor robustness** — drop each of the top-weighted donors and refit; if the estimate lurches when one donor leaves, the result hangs on a single market and is fragile.

These leave-one-out refits — and every robustness check from here on (placebo, spillover, launch-date sweep) — use **Step 0’s classical SLSQP fitter**: deterministic, no MCMC, milliseconds per refit, and a *point estimate only*. We benchmark each dropped-donor bar against that same fitter’s own all-donor baseline (the grey line, i.e. Step 0’s estimate) so the comparison is apples-to-apples; that SLSQP baseline runs a touch tighter than the Bayesian fit because the Dirichlet prior lightly regularizes the blend — which is why the plot labels “all donors (SLSQP)” and “all donors (Bayesian)” as two nearby-but-distinct numbers.

pre-fit gate: pre-RMSE €2.86k vs $|weekly\ lift|/3 = €5.1k \rightarrow$ PASS
pre-launch gap mean €0.30k (~0 = good fit). L00 total range €309–337k vs the same-fitter SLSQP baseline €318k (Bayesian €305k) – no single donor drives it.



Across 24 fresh panels – recovery bias €+23k (+7%), with a LARGE seed-to-seed sd of €59k.

PER-WEEK band covers the true path 86% of the time – ~ nominal; the posterior-predictive obs-noise term earns this.

But the TOTAL-lift 90% interval covers only ~50% across panels: SC’s sample-to-sample variance dwarfs any single panel’s posterior width, so the total interval is optimistic.

=> the CALIBRATED inference is the placebo permutation test (below), not the posterior band alone.

Why does the total-lift band under-cover while the per-week bands are fine? This is the notebook's most important negative result, and it is not a mystery — it follows from one modelling choice. The likelihood treats the weekly errors as independent, so if e_t is the error of the synthetic blend in week t , the model prices the uncertainty of the $n = 20$ -week **total** as

$$\text{sd}\left(\sum_{t \in \text{post}} e_t\right) = \sqrt{n} \sigma \quad (\text{iid errors}).$$

But the blend's true error is *not* iid. Whatever part of the treated market's factor loadings the weights fail to match leaks the shared factors straight into the gap,

$$e_t \approx (\lambda_0 - \sum_j w_j \lambda_j)^\top (g_t, s_t, m_t) + \varepsilon_{0t},$$

and m_t is a **random walk** — so the mismatch term is a *persistent, slowly wandering offset*, not fresh noise each week. For correlated errors the variance of a sum picks up all the covariance terms,

$$\text{Var}\left(\sum_{t=1}^n e_t\right) = \sigma^2 \left(n + 2 \sum_{k=1}^{n-1} (n-k) \rho_k \right) \xrightarrow{\rho_k \rightarrow 1} n^2 \sigma^2 \gg n \sigma^2,$$

where ρ_k is the error's lag- k autocorrelation. **Per-week bands only need the marginal variance** σ^2 , which the posterior-predictive ε_t term gets roughly right — that is why they cover $\approx$ nominal. **The total needs the covariances too**, and the iid likelihood sets every ρ_k to zero. The cell below measures both faces of this on live data: the *in-sample* pre-period residual shows only modest, sign-flipping autocorrelation (the fit **whitens** — scrubs the serial correlation out of — most of what it can see in-sample, which is why eyeballing the pre-fit residual would *not* warn you), while the *out-of-sample* gap error — measured on placebo worlds where the true lift is zero, so the post gap **is** the error — is almost perfectly persistent from week to week.

posterior sd of the 20-week total (mean across panels): €24k vs actual seed-to-seed sd of the total error: €59k -> ~2.5x underestimated

in-sample pre-period residual ACF (lags 1-3): [-0.03 0.38 -0.12] - modest and sign-flipping (noise band ~ +/-0.32); the fit whitens most of what it can see out-of-sample gap-error autocorrelation about 0, placebo worlds (lags 1,2,3,5,8): [0.93 0.93 0.92 0.93 0.94]

Symptom: the total-lift band under-covers (above). Mechanism: the blend's error is a PERSISTENT offset - the unmatched share of the macro random walk - not fresh weekly noise, so the variance of the 20-week SUM scales like n^2 , not the n the iid likelihood charges for. Per-week bands only price the marginal variance, which is why they stay ~ nominal while the total band is optimistic. Lesson: iid likelihoods and aggregated-over-time estimands don't mix - for totals, use design-based placebo inference (Depth B) or a likelihood with correlated (AR / random-walk) errors.

5x · Point estimate vs posterior — what the Bayesian layer actually bought

Step 0 produced a lift estimate and a p-value with no likelihood, no prior and no sampler; §4 produced a posterior. They target the *same* estimand, so lay them side by side on it — the total post-launch lift — and then do the thing that actually settles which **uncertainty** statement to believe: re-run *both* intervals on the fresh panels of the multi-seed check above, and count how often each covers the planted truth. Nothing below is asserted; all of it is measured.

SAME estimand - total lift over the 20 post-launch weeks. Planted truth €339k.

	est	5%	95%	width	truth?
classical SC (SLSQP) * placebo interval	318	245	401	156	yes
Bayesian SC (Dirichlet) * posterior CrI	305	270	339	69	NO

LOCATION - they agree: the estimates differ by €13k (4% of the effect), and both sit BELOW the truth.

WIDTH - they do not. Three rival scales for the sd of the 20-week total:

posterior sd (iid likelihood, this panel)	€	21k	
placebo / design-based sd (this panel)	€	60k	
ACTUAL seed-to-seed sd of the error (§5)	€	59k	<- the referee

-> the posterior underprices it ~2.8x; the placebo null misses it by ~1%.

COVERAGE across the same 24 fresh panels (nominal 90%):

Bayesian total-lift 90% credible interval	50%	
classical 90% randomization interval	88%	<- design-based, and it keeps its promise

P(lift > €300k cost) - the number §6 signs off on:

classical	not defined	
Bayesian posterior (§4)	0.59	-> TEST FURTHER
same posterior, re-scaled to the design sd	0.53	-> TEST FURTHER

('not defined' is no dodge: a p-value ranks data against a null; it attaches no probability to a hypothesis about the effect.)
-> the call HOLDS under the honest scale - but the confidence attached to it was borrowed, not earned.

1 · On the number itself, they agree — and neither is a hero. The classical SLSQP fit and the Bayesian posterior mean land within a few percent of each other (printed above), and *both* undershoot the planted truth. If anything the classical point estimate is the closer of the two in this panel: the Dirichlet prior spreads weight over many donors, and a regularized blend is a slightly more conservative one. So the Bayesian layer did **not** buy a better point estimate here. It could not have: the causal work was done by §3's identification and by the simplex, both of which the two estimators share. The undershoot is a property of *synthetic control*, not of either apparatus (§7 keeps it as a caveat).

2 · On the uncertainty, they disagree — and the classical arm wins. This is the uncomfortable, important finding of this notebook, and it is measured, not asserted. Three numbers price the same object (the sd of the 20-week total): the posterior's, the placebo distribution's, and —

the referee — the *actual* seed-to-seed spread of the estimation error across fresh panels. Read them off the print-out. The posterior underprices the truth by a factor printed above; **the placebo null lands essentially on it**. §5 already diagnosed exactly why the posterior misses: the likelihood treats the weekly errors as **iid**, but the blend’s error is a **persistent offset** — the unmatched share of the macro random walk — so the variance of a 20-week *sum* grows like n^2 , not the n the iid likelihood charges for. Hence the coverage table: the Bayesian 90% interval on the total covers far below its nominal rate across panels, while the classical **randomization interval** — **built by refitting the same estimator on untreated markets** — **covers at close to its nominal 90%**.

Read that again, because it is the reverse of the cookbook’s usual story. The design-based interval never wrote a likelihood, so it had no likelihood to get wrong: it takes the *actual* distribution of the estimator’s error from the donors themselves, persistent macro mismatch and all, instead of assuming a shape for it. Here the classical arm’s uncertainty statement is simply **the more trustworthy one**, and the honest way to report this analysis is to lead with the placebo test, not with the credible interval — which is precisely what §7 and the memo do.

3 · What the posterior *does* buy — and the tension it comes with. Look at the last block of the print-out. $P(\text{lift} > \text{cost})$ exists in exactly one of the two columns. The classical row reads *not defined*, and that is the literal truth rather than a rhetorical concession: a p-value ranks the data against a null, it attaches no probability to a hypothesis, and no amount of placebo refitting will turn a rank into $P(\text{the campaign paid})$. But §6’s go/no-go rule is nothing *but* that probability — the €300k cost has to meet the whole distribution of the lift, not its endpoints — so the decision quantity lives only in the Bayesian arm.

And now say the uncomfortable part plainly: **in this notebook, the decision quantity lives in the arm whose uncertainty we just showed to be the least trustworthy**. $P(\text{lift} > \text{cost})$ is read off the same posterior whose total-lift interval under-covers, so it inherits that optimism directly: re-centre nothing, merely re-scale the posterior’s sd to the design-based one, and the probability moves (printed above). In *this* run the verdict survives the re-scaling — TEST FURTHER either way — but that is luck, not robustness: a run sitting near the 0.9 GO bar would have had its rollout decision overturned by the same correction, and nothing in the posterior’s own output would have warned you.

So the ledger, honestly. Bayes did **not** buy the point estimate (the classical one is at least as good), did **not** buy interpretable weights (the classical blend is sparser), and did **not** buy a trustworthy interval on the total (it bought a *worse* one). What it bought is real but narrower than the brochure: a near-nominal **per-week** band, where the iid likelihood is roughly right and the posterior predictive’s noise term does honest work; a coherent way to propagate that uncertainty into euros; and the **only** object in the notebook that can answer the question the business asked. The fix for the tension is already written down in §5’s lesson, and it is not “abandon Bayes” — it is *stop feeding a total to an iid likelihood*: either price the total with design-based placebo inference (as this section just did), or give the model correlated errors (an AR(1) or random-walk residual) so it can see the persistence itself. Use the posterior for the decision; use the placebos for the confidence in it — and if you only have budget for one sentence in the memo, make it the placebo one.

6 · Decide, in euros

The campaign cost **€300k**. Everything the notebook has produced so far now collapses into a single object: the **posterior over total incremental sales** across the 20 post-launch weeks (`total_post`, in €000). The rollout question is *not* “is the lift positive?” — Depth B’s placebo permutation answers that — but “**what is the probability the lift beats what we paid?**”, because a real-but-too-small effect still loses money at national scale.

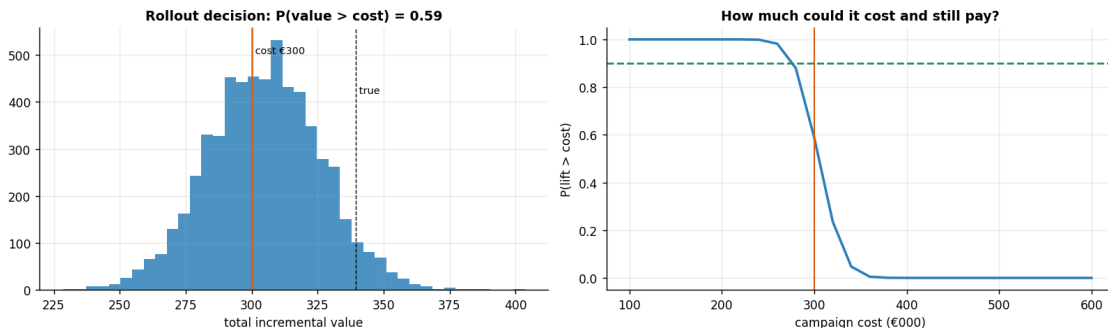
The decision rule (`policy.go_no_go`, the cookbook’s standard): read $P(\text{value} > \text{cost})$ off the posterior and act on a stated convention —

- $P > 0.9 \rightarrow$ **GO** — sure enough to act without buying more evidence;
- $P < 0.5 \rightarrow$ **NO-GO** — more likely than not the campaign doesn’t pay at this price;
- in between $\rightarrow$ **TEST FURTHER** — the honest middle where the effect may well be real but the *economics* are not yet settled.

Alongside the probability it reports the expected **net value** ($E[\text{value}] - \text{cost}$, in €000) and the expected **ROI** as a ratio ($\text{net} \div \text{cost}$, so break-even is 0) — three views of the same posterior, so nobody has to reverse-engineer the call from a chart.

How to read the two panels. *Left:* the posterior of total lift with the €300k cost line — the mass to the right of the line is $P(\text{lift} > \text{cost})$ (the dashed marker is the true total, which we only know because this is a simulation). *Right:* the same posterior read as a **headroom curve** — sweep a hypothetical cost c and plot $P(\text{lift} > c)$. Where the curve crosses the green 0.9 line is the *highest price at which this evidence would already justify a confident GO* — the number to bring to the media-buying negotiation, not just a yes/no.

```
{
  "P_value_gt_cost": 0.59,
  "expected_value": 304.77,
  "expected_net_value": 4.77,
  "expected_roi": 0.02,
  "value_lo": 270.24,
  "value_hi": 339.45,
  "cost": 300.0,
  "decision": "TEST FURTHER"
}
```



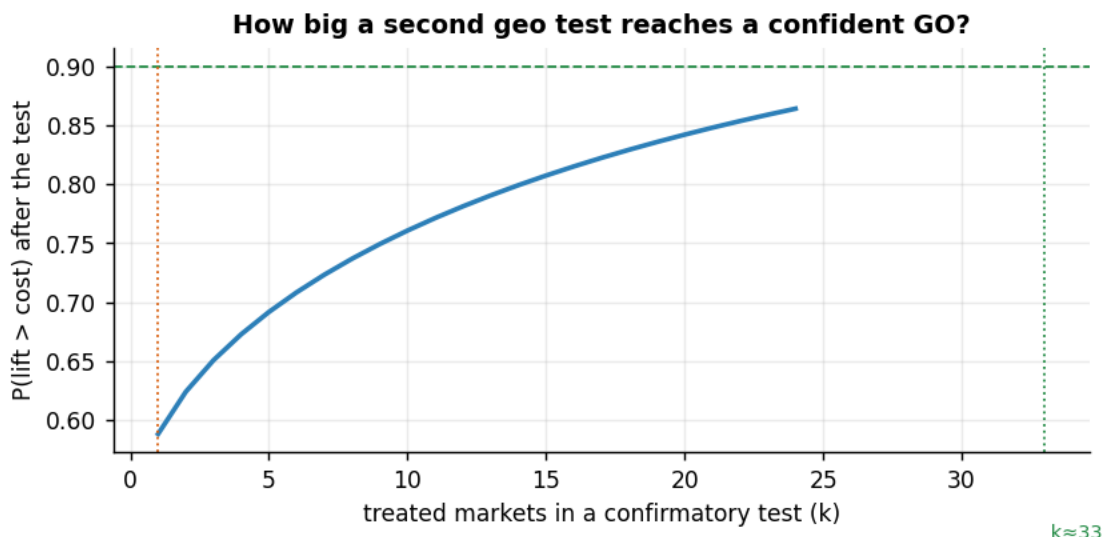
Read-out. The verdict in this run is **TEST FURTHER**, and the JSON says exactly why: the expected lift ($\approx$ €305k) sits almost on top of the €300k cost — expected net value $\approx$ **€5k**, ROI $\approx$ **+2%** — and $P(\text{lift} > \text{cost}) \approx 0.59$, barely better than a coin flip. Hold this against Depth B without flinching at the apparent contradiction: the placebo permutation says the effect is **real** ($p \approx 0.03\text{--}0.05$), while the economics say it is **marginal at this price**. Both are true at once — “statistically significant” and “worth the money” are different questions, and conflating them is how marginal campaigns get scaled. The headroom curve prices the gap: the 0.9 crossing sits at $\approx$ €275k in this run, so had the campaign cost $\sim 8\%$ less, this same evidence would already be a confident GO. At €300k, the evidence buys conviction that *something* happened, not conviction that it *paid*. And read that probability with 5x’s warning in hand: it is computed from the posterior whose total-lift interval under-covers, so the *precision* behind “0.59” is itself optimistic — which is one more reason the call below is “test further,” not “the number says go.”

A borderline call has a price tag: the value of information. “TEST FURTHER” must not end as a shrug — the deliverable is an answer to “*what evidence would settle this, and is it worth buying?*” Two tools, both computed below:

- **VOI (value of information).** $\text{VOI} = \mathbb{E}[\max(V - c, 0)] - \max(\mathbb{E}[V] - c, 0)$ — the expected gain from deciding *after* the uncertainty resolves versus committing now on the mean. Intuition: if the truth lands below the cost line, perfect foresight would have let us walk away and keep the difference; VOI is that averted loss, averaged over the posterior. It is the most a *perfectly informative* test could add in expected decision value — i.e. an **upper bound on any confirmatory-test budget**.
- **Test sizing.** A second geo test treating k independent markets tightens the posterior roughly as $1/\sqrt{k}$ — it **sharpens** the estimate around the same mean; it doesn’t move it. So “how big must the test be?” becomes: *the smallest k whose tightened posterior reaches $P(\text{lift} > \text{cost}) \geq 0.9$* . Two hidden assumptions live inside that $1/\sqrt{k}$, and both flatter the test: it treats the k market-level estimates as **independent** with equal variance — but this whole world is built on *shared* macro factors, so the k gap estimates share estimation error and the *effective* k is smaller than the nominal one — and it takes this posterior’s sd as the per-market scale, which the multi-seed check just showed runs optimistic. Both push the same way: read the computed k as a **lower bound**.

Value of information: resolving the cost-line uncertainty is worth $\sim$ €6k of expected decision value.

Reaching $P(\text{lift} > \text{cost}) \geq 0.90$ would take ~ 33 independent treated markets - more than this world contains (30 DMAs, one already treated). The honest read: NO feasible single confirmatory test gets to 0.90 here; extend the post-period, pool repeated tests, or accept a lower confidence bar - and the multi-seed check says even this arithmetic is optimistic. Still an experiment, never a blind launch.



Read-out — sizing the confirmatory test. $\text{VOI} \approx \text{€}6\text{k}$ in this run: with the posterior centred almost exactly on the cost line, even perfect information barely changes the expected outcome — whichever way the truth falls, we’re near break-even, so the *stakes of deciding wrong are small*. That immediately disciplines the next step: a confirmatory test is only rational if it costs less than $\sim \text{€}6\text{k}$ of all-in decision value — no $\text{€}50\text{k}$ research program is justified by these economics. Now the sizing arithmetic, read straight off the cell above: reaching the 0.9 bar takes **the number of independent treated markets printed there**, and because the multi-seed check in step 5 showed the interval is *optimistic*, treat that count as a *lower* bound. Whether that many markets is even available — this world holds 30 DMAs, one already treated — is what decides the branch: if the required count fits inside the pool, **that confirmatory geo test is the experiment to run**; if it exceeds the pool, **no single geo test settles the question at this price**, and the fallbacks below take over. Those honest fallbacks hold either way: **renegotiate the price** (at $\approx \text{€}275\text{k}$ the existing evidence already clears 0.9), **extend the post-period or pool repeated tests** (more weeks per market buys precision too), or **accept a lower confidence bar** knowing the multi-seed check says these intervals are already optimistic. What is *not* on the list: a blind national rollout. Depth C pulls this thread together with the launch-date and donor-pool sensitivities into the one-paragraph verdict a VP signs.

7 · Caveats

- **Here synthetic control is high-variance, not conservative.** In this panel the point estimate lands $\sim 10\%$ low, but across seeds (Step 5) it shows only a small positive mean bias (single digits to low-teens % across seeds) but a *large* sample-to-sample spread — and its posterior interval is *optimistic* about the total lift. So trust the **placebo permutation test** (Step 0’s classical inference), not the point estimate or the band — 5x puts the two intervals on the same panels and measures their coverage, and it is the design-based one that keeps its 90% promise — §5 shows the mechanism: the blend’s error is a *persistent* (serially correlated) offset, so the iid likelihood underprices the variance of any multi-week total.
- **Pre-fit is a gate, not a formality.** High pre-RMSE $\Rightarrow$ the post gap is uninterpretable; no

Bayesian machinery fixes a bad donor pool.

- **No-anticipation and no-spillover are assumptions.** Anticipation is stress-tested by placebo-in-time in Depth B; spillover is *invisible to every placebo*, so Depth B stress-tests it by simulation instead — planted spillover **attenuates** the estimate (SC under-states, never inflates), and the fix is donor-pool design, not statistics.
- **One treated unit = limited power.** Significance comes from the permutation test, coarse with few donors.

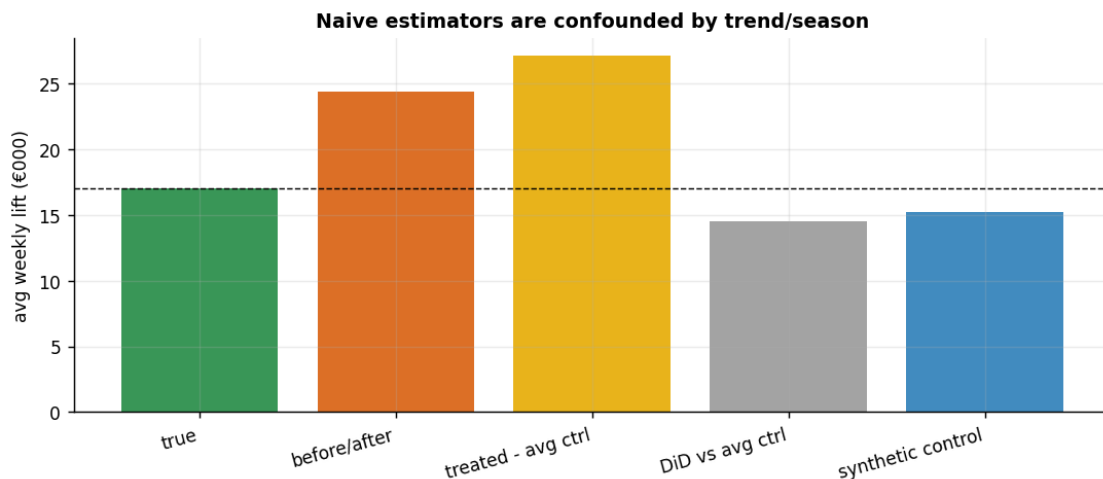
Depth A · Why the naive estimators fail

Two tempting shortcuts, both confounded by the shared trend/season/macro wave:

- **Before/after:** post mean — pre mean of the treated DMA — books the entire trend as effect.
- **Treated — average control:** the average control has the wrong factor loadings, so it doesn't cancel the common wave cleanly.

Synthetic control is the fix: a *weighted* control chosen to match pre-launch.

```
{  
  "true": 17.0,  
  "before/after": 24.3,  
  "treated - avg ctrl": 27.1,  
  "DiD vs avg ctrl": 14.5,  
  "synthetic control": 15.2  
}
```



Read-out. Before/after badly overstates the lift (it books the trend); treated-minus-average is **worse** — it books the raw between-market level gap, differencing nothing out; differencing out the shared trend (DiD vs the average control) removes the trend bias yet a *level* gap from mismatched factor loadings remains; **the reweighted synthetic control comes closest** (€15.2k vs the true

€17.0k — about a 10% undershoot; DiD-vs-average lands at €14.5k, and neither naive shortcut is even close). Same data, four estimates — the estimator *is* the identification strategy.

Depth B · Placebo inference — is the gap statistically real?

Step 0 already ran this test and printed its p-value; it is the classical arm's *entire* inferential apparatus, so it deserves the box opened. Recall the question, which is a **permutation** question rather than a standard-error one (with one treated market there is no standard error to have):

If the campaign had no effect, how unusual would a post-launch gap this big be, just by chance?

Placebo-in-space, drawn out. Pretending in turn that *each donor market* was the treated one gives a whole family of gap paths — none of them from a market that actually ran a campaign — and the left panel plots them all: the grey spaghetti is the null, the coloured line is our metro. The right panel is the same comparison on Abadie's scale-free statistic (post/pre RMSE ratio; Step 0 defines both refinements — the pre-fit filter and the ratio — and why each is needed). Two things worth checking that a single p-value hides, and that this cell prints:

- **How many donors survived the pre-fit filter**, and therefore how coarse the reference distribution is. With ~30 markets the p-value is *granular*: it cannot resolve below roughly 1/30 no matter how strong the effect is. One treated unit buys limited power, and no amount of Bayesian machinery would buy more.
- **Whether the verdict depends on the filter cutoff** (2×, 5×, 20× the treated unit's pre-RMSE). A test whose answer moves with an arbitrary threshold is not evidence; this one does not.

placebo-in-space kept 18/29 donors (dropped 11: pre-RMSE > 5x the treated fit)

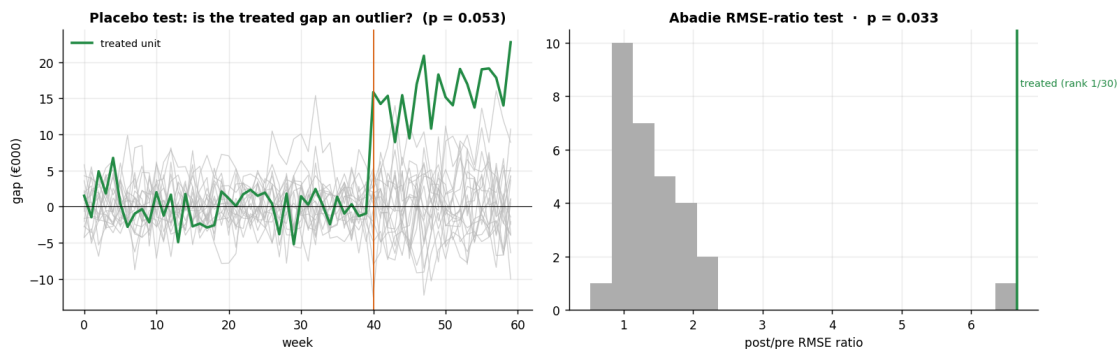
placebo p vs filter cutoff: 2x -> p=0.056 * 5x -> p=0.053 * 20x -> p=0.333

(tighter filters compare only well-matched placebos and sharpen the test; a loose 20x filter admits poor synthetics and dilutes it)

placebo p (gap) = 0.053 [add-one permutation, treated excluded from placebos] *

RMSE-ratio p = 0.033 [Abadie rank/N, treated included] (treated ratio 6.7 vs

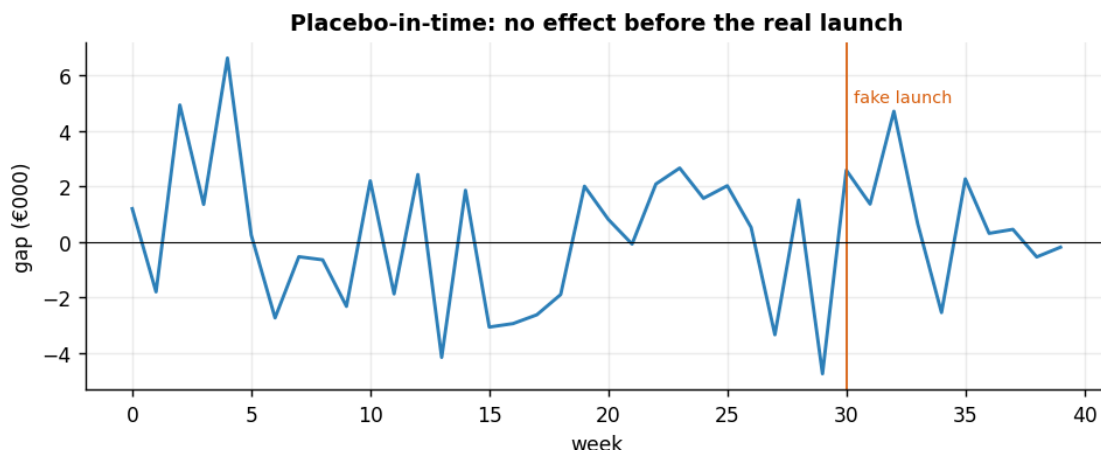
placebo median 1.3; both rank the treated 1st)



Two p-values, reconciled. The cell prints two significance numbers, and they differ by about two hundredths (0.053 vs 0.033) because they use slightly different bookkeeping. The **raw-gap permutation** p is an *add-one* permutation — the treated market counts as one draw of its own null and is *excluded* from the reference placebos — while **Abadie’s RMSE-ratio** p is a rank statistic that keeps the treated market *in* the denominator and divides its post-period fit by its pre-period fit. Both rank the treated market **1st of all units**, so the qualitative verdict is identical. We lead with the **RMSE-ratio**: it is scale-free and penalizes placebos that simply fit badly *everywhere*, so a poor-pre-fit donor can’t masquerade as a sharp post-launch jump.

Placebo-in-time. A second falsification: pretend the campaign launched *before* it did (a fake week inside the pre-period). The estimated “effect” should be ≈ 0 — a non-zero fake effect would mean the method invents lift (anticipation or overfitting).

placebo-in-time avg gap €0.9k vs pre-fit RMSE €2.86k (should be ~ 0 , and it is well within the pre-fit noise) - no spurious pre-effect.



Spillover — stress-testing the one assumption the placebos cannot touch. The assumption table in §3 was honest that placebo-in-space *cannot* detect spillover: if the campaign leaks into donor markets, every placebo refit is contaminated in exactly the same way, so nothing looks unusual. But the simulator can do what no real dataset can — rerun the same world with spillover *planted*. Below, a fraction φ of the true lift Δ_t leaks into five “neighbouring” donor markets ($Y_{jt} += \varphi \Delta_t$ for j in the neighbour set) and we refit. First-order prediction *before* running it: the contaminated donors rise, so the synthetic counterfactual rises, so the measured gap **shrinks** by roughly the spilled weight,

$$\text{bias} \approx -\varphi \sum_{j \in \text{neighbours}} w_j \times \text{lift},$$

i.e. spillover makes synthetic control **under-state** a real campaign — attenuation toward “no effect”, the opposite of the false-positive failure most people fear.

phi	est_total_k	weight_on_neighbours
0.00	317.74	0.15

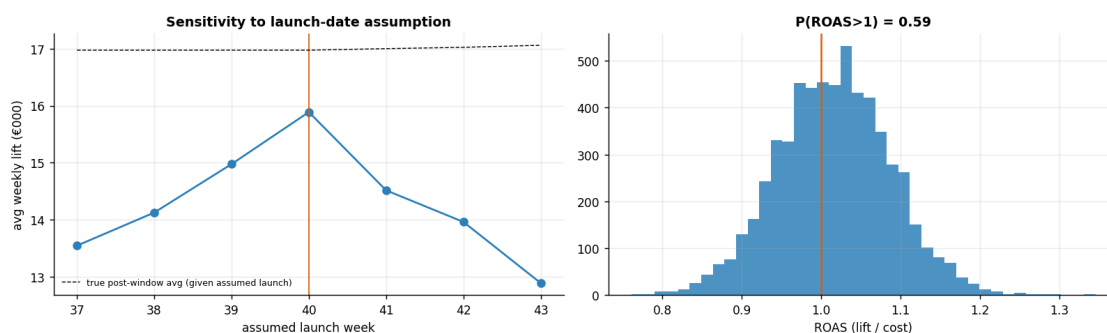
0.10	312.55	0.15	
0.25	304.75	0.15	
0.50	291.75	0.15	(true total €339k)

attenuation at $\phi=0.5$: measured €-26k vs first-order prediction $-\phi \cdot (\text{Sigma neighbour weights}) \cdot \text{lift} = \text{€-24k}$
excluding the 5 neighbours at $\phi=0.5$ restores the estimate to €311k
(contaminated pool: €292k; clean pool: €318k)
Read-out: spillover into donors raises the synthetic, so SC UNDER-states the lift - attenuation, not inflation. Conservative for a GO call, but it can turn a true GO into a NO-GO. And because no placebo can flag it, the fix is DESIGN, not statistics: exclude adjacent / media-overlapping donors up front - as just shown, a clean-but-smaller pool beats a bigger contaminated one.

Depth C · The euro rollout decision & sensitivity

Pull it together into the number a VP signs off on, and pressure-test how the call moves with the **launch-timing** assumption (did we mis-date the campaign?) and the **donor pool**.

```
{
  "true_total": 339.48,
  "estimated_total": 304.77,
  "ci90": [
    270.24,
    339.45
  ],
  "pre_rmse": 2.86,
  "p_space": 0.05,
  "p_rmse_ratio": 0.03,
  "placebo_in_time": 0.9,
  "P_lift_gt_cost": 0.59,
  "decision": "TEST FURTHER",
  "eff_donors": 14.58
}
```



The one-paragraph decision

The campaign drove a **real** incremental lift: the synthetic tracks the treated market closely before launch (low pre-RMSE, pre-gap ≈ 0), spreads its weight across ~ 15 effective donors (no single market dominates), and it is robust to dropping any single donor (leave-one-out stays close to the same-fitter baseline). Nudging the *assumed* launch date does shift the estimate — but that is mechanical, not fragility: mis-dating folds no-effect weeks into the post window and dilutes the average lift, so the peak at the true launch week is itself a diagnostic. The post gap is an outlier on both the raw-gap and the scale-free RMSE-ratio **placebo permutation** tests, and the time placebo shows no spurious pre-effect — that permutation test is the calibrated evidence the effect is real. The euro call is deliberately cautious: the point estimate carries meaningful seed-to-seed variance (this panel lands $\sim 10\%$ below truth; across panels the mean bias is small (single digits to low-teens %) but the variance is *large*), and the multi-seed check shows the **total-lift interval is optimistic** — so we do **not** let a single posterior interval authorise a national rollout. Against the €300k spend the effect sits near break-even: **“promising, confirm with a second geo test,” not “scale nationally today.”** The effect is clearly real; whether it clears *this* cost is the close call — and one an optimistic interval should make us *more*, not less, cautious about.

Method summary. Step 0’s classical read — Abadie’s SLSQP simplex weights plus placebo/permutation inference — reaches essentially the same lift as the Bayesian fit, and (see 5x) a *better-calibrated* statement of uncertainty about the total, because it never assumed an iid likelihood it could get wrong. The Bayesian layer earns its keep at exactly one place, the **decision**: only a posterior can return $P(\text{lift} > \text{cost})$. The discipline this notebook demands is to use each for what it is good at — the placebos for the confidence, the posterior for the call — and to remember that the euro probability inherits the optimism 5x measured. **On real data:** notebook **07b** runs this same design on a real Google geo experiment; Abadie’s California Proposition 99 study remains the canonical public panel to practise on.